



Insurance Premium Model

Multi-Factor Flood Insurance Premium Calculation

MKM Research Labs

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David K Kelly

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1 Executive Summary

The Insurance Premium Model calculates flood insurance premiums using a deterministic multi-factor approach. A base rate per £1,000 of property value is multiplied by seven factor loadings drawn from configuration tables, each representing a distinct risk dimension.

The premium formula is:

$$P = \frac{V}{1000} \times r_{\text{base}} \times f_{\text{type}} \times f_{\text{flood}} \times f_{\text{age}} \times f_{\text{construction}} \times f_{\text{area}} \times f_{\text{security}} \times f_{\text{claims}} + P_{\text{contents}}$$

where V is the property value, r_{base} is the base rate, and each f is a multiplicative factor from the corresponding lookup table.

The final premium is clamped to configurable bounds (`MIN_PREMIUM`, `MAX_PREMIUM`) to ensure commercial viability and regulatory compliance.

Key risk: Factors are independent with no interaction terms. Cross-effects (e.g. flood zone $\times$ construction type) are not captured.

2 Model Purpose and Scope

2.1 Purpose

The model serves two primary functions:

- **Premium quotation:** Compute annual flood insurance premiums for individual residential properties.
- **Portfolio pricing:** Support bulk premium calculation across property portfolios for underwriting and reserving.

2.2 Scope

The model covers UK residential properties with flood exposure. Factor tables are calibrated to UK market conditions and EA flood zone classifications.

3 Methodology

3.1 Factor Structure

Seven independent multiplicative factors adjust the base premium:

Table 1: Premium factor dimensions.

Factor	Input	Source Table
Property type	Detached / Semi / Terrace / Flat	PROPERTY_TYPE_PREMIUM_FACTORS
Flood risk zone	EA flood zone classification	FLOOD_RISK_PREMIUM_FACTORS
Building age	Construction year $\rightarrow$ band	AGE_PREMIUM_FACTORS
Construction type	Brick / Timber / Concrete / Steel	CONSTRUCTION_TYPE_FACTORS
Floor area	m^2 banded	AREA_PREMIUM_BANDS
Security	Alarm / locks / flood gates	Fixed discount
Claims history	Prior claims count	Loading multiplier

3.2 Age Banding

Properties are classified into four age bands based on construction year:

- **New build:** age < 10 years
- **Modern:** $10 \leq \text{age} < 30$ years
- **Established:** $30 \leq \text{age} < 100$ years
- **Heritage:** age ≥ 100 years

Each band maps to a factor range $(f_{\min}, f_{\max})$ from configuration.

3.3 Premium Bounds

The calculated premium is clamped:

$$P_{\text{final}} = \max(P_{\min}, \min(P_{\max}, P_{\text{raw}}))$$

ensuring that no premium falls below the minimum viable level or exceeds the regulatory cap.

4 Inputs and Outputs

4.1 Inputs

- Property value (£).
- Base rate per £1,000 (from `BASE_RATE_RANGE`).
- Property type, flood zone, construction year, construction type, floor area.
- Security features (boolean flags).
- Claims history count.
- Contents premium add-on (optional).

4.2 Outputs

- Annual insurance premium (£), clamped to bounds.

5 Assumptions and Limitations

5.1 Assumptions

1. All premium factors are independent (no interaction effects).
2. Factor tables are calibrated to current UK market conditions.
3. Property value is an accurate reflection of rebuilding cost.
4. EA flood zone classifications are current and accurate.

5.2 Limitations

1. No cross-factor interactions (e.g. flood zone $\times$ age).
2. Factor tables are static and require periodic recalibration.
3. No explicit modelling of climate change trends on flood premiums.
4. Contents premium is an additive flat amount, not risk-adjusted.

6 Sensitivity Analysis

This section analyses the sensitivity of the insurance premium to each factor.

7 Validation Questions

7.1 Q1 — Purpose

“Is the model being used for its intended purpose?”

Yes. The model calculates flood insurance premiums for UK residential properties, consistent with its design scope.

7.2 Q2 — Conceptual Soundness

“Is the model conceptually sound?”

The multiplicative factor approach is standard actuarial practice for property insurance rating. Factor independence is a simplification but is well-understood.

7.3 Q3 — Data Quality

“Is the data quality adequate?”

Factor tables are derived from industry data and EA classifications. Property data quality depends on upstream data feeds.

7.4 Q4 — Developmental Evidence

“Is there adequate developmental evidence?”

Factor tables are sourced from published industry rating guides. Unit tests verify calculation correctness and boundary clamping.

7.5 Q5 — Outcomes Analysis

“Are outcomes reasonable?”

Premium outputs are validated against market benchmarks for representative property archetypes.

7.6 Q6 — Ongoing Monitoring

“What ongoing monitoring is required?”

Annual review of factor tables against market loss experience. Quarterly monitoring of premium distribution statistics.

7.7 Q7 — Benchmarking

“Has the model been benchmarked?”

Premiums are compared against Flood Re pricing and commercial insurance quotes for representative properties.

7.8 Q8 — Monitoring

“What type of monitoring is required?”

Monitor premium adequacy ratios (premium / expected loss) by flood zone. Track claims frequency and severity against premium assumptions.

7.9 Q9 — Overall Risk

“Is the overall model risk acceptable?”

The model is a standard actuarial rating tool. Risk is Low provided factor tables are recalibrated annually.